INTERNSHIP ASSESSMENT REPORT – CIE 3

CHAPTER - 03: ROLES AND RESPONSIBILITIES DURING OJT-3

3.1 Introduction

During my On-the-Job Training (OJT-2), I worked on developing an AI-based Interview Analyzer System**.** The objective of this project was to simulate a real interview environment and evaluate candidate responses using Artificial Intelligence techniques.

The system allows users to log in, select a stream, and choose between text or voice mode to answer interview questions. It includes both technical and HR questions and evaluates answers using advanced Natural Language Processing (NLP) techniques. The application was developed using Python and Flask for backend, and HTML, CSS, and JavaScript for frontend.

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3.2 Ability to Apply Skills & Technical Knowledge

During this project, I effectively applied my knowledge of Artificial Intelligence, Machine Learning, and Natural Language Processing. I implemented semantic answer evaluation using a pre-trained model from Sentence Transformers,

which converts text into embeddings. These embeddings are compared using cosine similarity to calculate how closely the user's answer matches the ideal answer.

I used Flask to handle backend logic such as question flow, session management, and result calculation. On the frontend, I implemented voice input using browser-based speech recognition and designed an interactive interface.

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3.3 Intern's Performance on Assigned Tasks and Project

During my internship, I successfully completed all assigned tasks related to the development of the AI Interview Analyzer.

I developed the complete workflow including login system, stream selection, interview interface, and result dashboard. The system dynamically generates questions and maintains proper session flow.

I implemented both text and voice modes, where coding questions are filtered out in voice mode to improve usability. I also integrated semantic scoring using AI models and tested the system with multiple inputs to ensure accuracy.

Additionally, I debugged issues related to question display, scoring logic, and session handling. Overall, I completed the project efficiently and demonstrated consistent performance.

3.4 Value Addition to the Organization

The AI Interview Analyzer adds value by providing an automated and intelligent solution for evaluating candidates during interviews.

The system simulates a real interview environment and provides instant feedback including scores, confidence percentage, strengths, weaknesses, and suggestions. This reduces manual effort and improves evaluation efficiency.

It can be used in educational institutions for student practice, in companies for initial screening, and in training platforms. The project demonstrates how AI can enhance traditional interview processes and improve decision-making.

3.5 Roles and Responsibilities

- Developed the AI Interview Analyzer system using Python and Flask.
- Designed frontend pages such as login, stream selection, interview interface, and result dashboard using HTML and CSS.
- Implemented both text and voice input modes using JavaScript speech recognition.
- Integrated AI-based semantic scoring using Sentence Transformers and cosine similarity.
- Developed logic for dynamic question generation and filtering coding questions in voice mode
- Implemented result analysis including scores, confidence percentage, and feedback generator.
- Added anti-cheating features such as tab switch detection and disabling copy-paste.
- Tested and debugged the application to ensure smooth performance and accuracy.

3.6 Application of Technical Knowledge

In this project, I applied my technical knowledge to build a complete AI-driven interview evaluation system.

User answers were processed using NLP techniques, where both user input and ideal answers were converted into embeddings using a pre-trained model. These embeddings were compared using cosine similarity to generate a meaningful score based on semantic similarity.

The system calculates technical and HR performance separately and generates an overall confidence score. Based on these results, structured feedback such as strengths, weaknesses, and suggestions is provided.

This implementation helped me understand how theoretical AI concepts are applied in real-world systems.

3.7 Learning Outcomes

During this project, I gained practical experience in developing AI-based applications using Natural Language Processing.

I learned how to use pre-trained models for semantic analysis and integrate them into a web application. I improved my skills in Python, Flask, and frontend technologies.

I also gained experience in handling real-time user input, session management, and debugging. This project enhanced my problem-solving skills and confidence in working on real-time AI systems.

3.8 Personal Reflection

During the development of this project, I initially faced challenges in implementing accurate answer evaluation and managing dynamic question flow.

However, through continuous learning and experimentation, I was able to overcome these challenges. This project improved my understanding of AI models and strengthened my ability to build real-time applications.

It also helped me realize the importance of user experience, system design, and testing in software development.

3.9 Future Scope

Future enhancements of this system include real-time proctoring, camera-based monitoring, and advanced AI-based feedback generation.

The system can also be improved by training custom models for better accuracy and integrating real-time analytics for deeper performance insights.

3.10 Conclusion

In conclusion, the AI Interview Analyzer successfully demonstrates the application of Artificial Intelligence and Natural Language Processing in interview evaluation.

The system provides an efficient and automated way to conduct and analyze interviews, making it useful in both educational and professional environments. Through this project, I applied my technical knowledge, completed assigned tasks effectively, and developed a solution with real-world impact.

3.11 USE CASE: AI INTERVIEW ANALYZER SYSTEM

3.11.1 Introduction

This use case describes the application of an AI-based Interview Analyzer system that evaluates candidate responses during interviews using Artificial Intelligence and Natural Language Processing (NLP). The system is designed to simulate a real interview environment where users can answer questions in either text or voice mode. It includes both technical and HR questions and provides instant evaluation based on semantic similarity.

The system analyzes user responses, generates scores, calculates overall confidence, and provides structured feedback such as strengths, weaknesses, and suggestions. This makes the system useful for interview preparation, training, and evaluation purposes.

3.11.2 Problem Statement

In traditional interview processes, evaluation is done manually by interviewers, which can be time-consuming and sometimes inconsistent. It is difficult to evaluate answers accurately based on meaning, as most systems rely on keyword matching instead of understanding context. Candidates also do not receive immediate feedback, which limits their ability to improve.

Additionally, in online interview environments, maintaining fairness and preventing cheating is a challenge. Therefore, there is a need for an automated system that can evaluate answers intelligently, provide instant feedback, and simulate a real interview experience.

3.11.3 Proposed Solution

To address these challenges, an AI-based Interview Analyzer system is developed. The system uses NLP techniques to evaluate answers using semantic similarity rather than keyword matching. It provides automated scoring, confidence analysis, and structured feedback.

Voice mode is included for realistic interaction, and coding questions are filtered for usability. Basic anti-cheating mechanisms are also implemented.

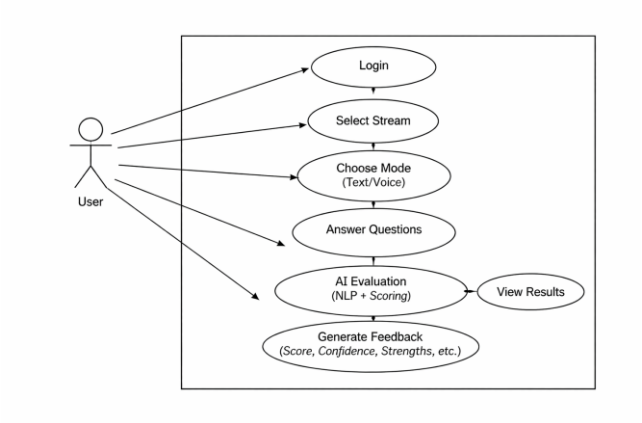
3.11.4 Workflow of the AI Interview Analyzer

1. User logs into the system using credentials.
2. User selects the desired stream (e.g., AI/ML, Data Science, etc.).
3. User chooses interview mode (text or voice).
4. The system dynamically generates a set of technical and HR questions.
5. Questions are displayed one by one to simulate a real interview.
6. The system processes answers using NLP techniques.
7. Both user answer and ideal answer are converted into embeddings using a pre-trained model.
8. Cosine similarity is calculated to determine semantic similarity between answers.
9. Scores are generated for each question.
10. Final results include individual scores, performance analysis, strengths, weaknesses, and suggestions.

3.11.5 Use Case Diagram

The following diagram represents the interaction between the user and the AI Interview Analyzer System:

Figure 3.1: AI Interview Analyzer Workflow



3.11.6 Output Screenshots

Figure 3.2: Login Page of AI Interview Analyzer

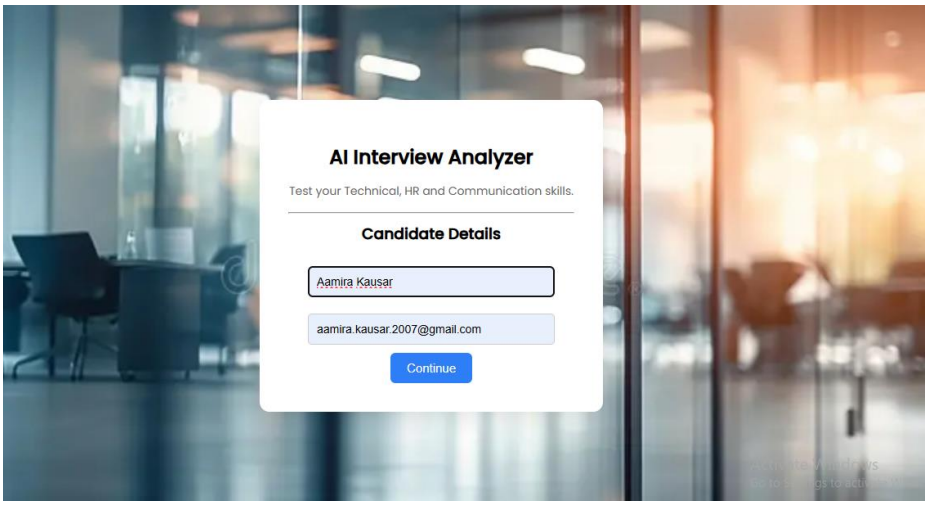


Figure 3.3: Stream and Mode Selection

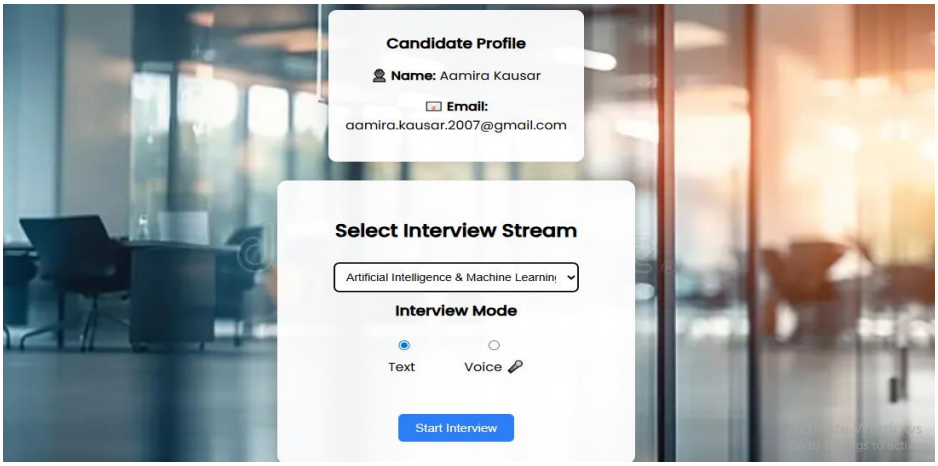
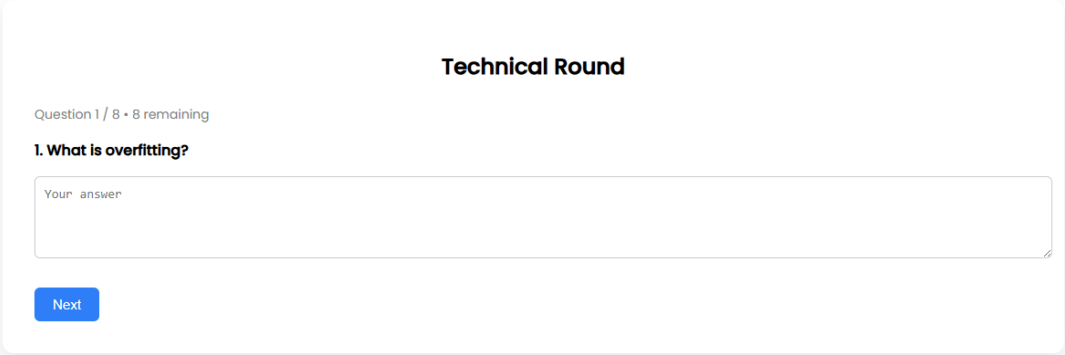


Figure 3.4: Question Panel (TECHNICAL & HR rounds)



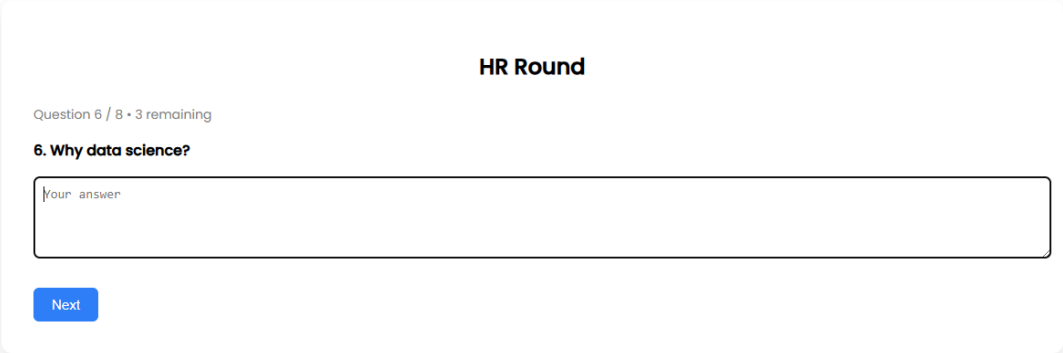
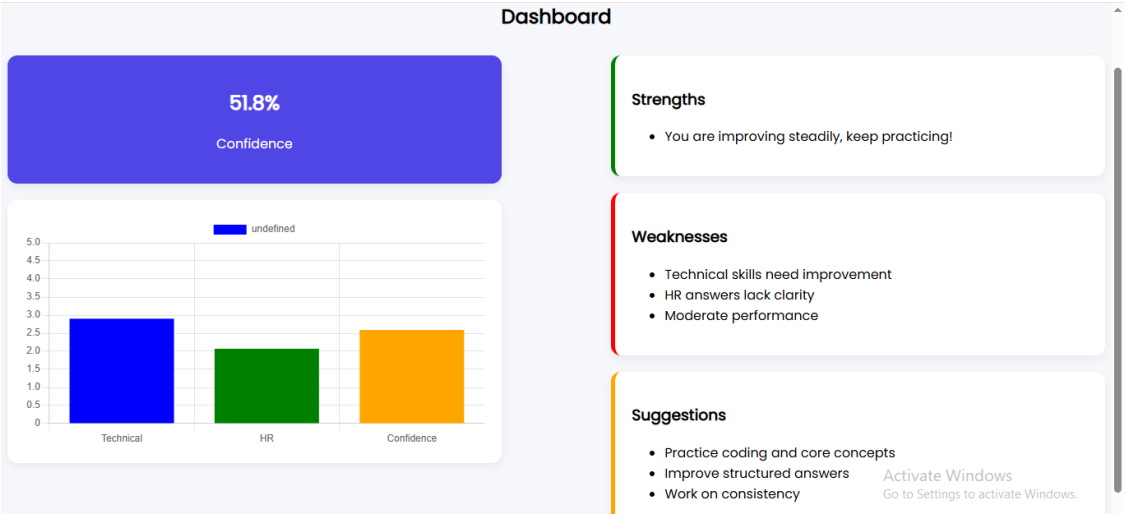


Figure 3.5: Result Page with Strengths, Weaknesses & Suggestions



3.11.7 Benefits of the System

- Provides automated and unbiased evaluation.
- Reduces time and effort in interview assessment.
- Improves candidate performance through instant feedback.
- Simulates a real interview environment.
- Can be used in education, recruitment, and training platforms.

3.11.8 Conclusion

The AI Interview Analyzer use case demonstrates how Artificial Intelligence and Natural Language Processing can be used to improve interview evaluation systems.

The system provides an efficient, scalable, and intelligent solution for assessing candidate responses and enhancing interview preparation. It highlights the practical application of AI in real-world scenarios and shows how traditional processes can be improved using technology.